PATUAKHALI SCIENCE AND TECHNOLOGY UNIVERSITY

COURSE CODE EEE-212

Electrical Technology Sessional

SUBMITTED TO:

Md. Naimur Rahman

Professor
Department of Electrical and Electronics Engineering
Faculty of Computer Science and Engineering

SUBMITTED BY:

Hridoy Chandra Sarker

ID: **2102019**,
Registration No: **10146**
Faculty of Computer Science and Engineering

Seemanta Shill

ID: **2102025**,
Registration No: **10152**
Faculty of Computer Science and Engineering

Date of submission:
Project title: **Smart Plant Monitoring System**

Project Proposal: Smart Plant Monitoring System

1. Project Title:
Smart Plant Monitoring System using NodeMCU ESP8266

2. Introduction:
The Smart Plant Monitoring System automates and optimizes plant care by monitoring environmental factors such as soil moisture, temperature, and humidity. The system uses sensors and IoT technology to provide real-time feedback and automatically water plants when needed, minimizing manual intervention.

3. Problem Statement:
Many people face challenges in maintaining healthy plants due to a lack of knowledge or time. Key environmental factors such as soil moisture, temperature, and humidity need to be consistently monitored to ensure plant health. A smart system that automatically tracks and responds to these factors can greatly improve plant care and reduce the risk of neglect.

4. Objectives:

- To design a smart monitoring system that measures soil moisture, temperature, and humidity, and automates plant watering based on sensor inputs.
- To enable remote monitoring via Wi-Fi, giving users real-time updates on the plant's environment.
- To display environmental data on an LCD screen and provide manual control with a push button.

5. Proposed Solution:
The Smart Plant Monitoring System will use the DHT11 sensor to monitor temperature and humidity, along with a soil moisture sensor to track water levels in the soil. A relay module will control a water pump to irrigate the plant when necessary. The NodeMCU ESP8266 microcontroller will handle data processing and transmit the information via Wi-Fi, allowing users to monitor plant conditions remotely. Additionally, the system will display real-time sensor data on an LCD screen and allow manual control via a tactile push button.

6. Components:

- **NodeMCU ESP8266:** Microcontroller with Wi-Fi for remote connectivity.
- **Soil Moisture Sensor:** Measures the moisture content in the soil.
- **DHT11 Temperature and Humidity Sensor:** Monitors ambient temperature and humidity levels.
- **PIR Motion Sensor:** Detects motion near the plant for additional monitoring or security.
- **Relay Module:** Controls the water pump based on sensor readings.
- **Water Pump:** Automates irrigation when soil moisture is low.
- **LCD Display:** Shows real-time sensor data for local monitoring.
- **Breadboard and Jumpers:** For connecting components.
- **18650 Battery:** Powers the system.
- **Tactile Push Button:** Allows manual activation of the water pump.

7. Methodology:

- **Hardware Setup:**
  Sensors will be connected to the NodeMCU ESP8266. The DHT11 sensor will monitor the temperature and humidity, while the soil moisture sensor tracks the soil's water content. A relay module will activate the water pump when the soil is dry. The system will be powered by a 18650 battery, and data will be displayed on an LCD screen for local monitoring.
- **Software Development:**
  The NodeMCU will be programmed to read data from the DHT11, soil moisture sensor, and PIR motion sensor. Based on predefined conditions, the microcontroller will trigger the water pump and send real-time data via Wi-Fi for remote monitoring. The system will feature a user-friendly web or mobile interface for easy access to data and control.

8. Expected Outcomes:

- Accurate monitoring of soil moisture, temperature, and humidity.
- Automated irrigation based on soil moisture levels.
- Real-time remote monitoring of environmental factors.
- Display of data on an LCD screen for on-site viewing.

9. Timeline:

- **Week 1-2:** Component selection and research.
- **Week 3:** Hardware assembly and initial testing.
- **Week 4:** Software development for sensor integration and Wi-Fi communication.
- **Week 5:** Final system integration and testing.
- **Week 6:** Project completion and presentation.

10. Budget:

- NodeMCU ESP8266: 420 Tk
- Soil Moisture Sensor: 120 Tk
- DHT11 Temperature and Humidity Sensor: 185 Tk
- PIR Motion Sensor: 110 Tk
- Relay Module: 95Tk
- Water Pump: 160 Tk
- LCD Display: 220 Tk
- Breadboard and Jumpers: 180 Tk
- 18650 Battery: 170 Tk
- Tactile Push Button: 7 Tk
- Others: 260 Tk

  Total: 1927 Tk

11. Conclusion:
This Smart Plant Monitoring System, enhanced with temperature and humidity monitoring through the DHT11 sensor, provides a comprehensive solution for plant care. The integration of multiple environmental sensors ensures optimal plant health by automating the watering process and providing real-time feedback to users via a remote interface.